Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–September 11th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Which of the PRAM model's assumptions help to compute the time complexity? (1)

B. Which of the assumptions of the PRAM model simplifies the communication between processor and memory? (1)

C. Please define when a write conflict occurs in a PRAM model. (1)

D. Define the "cost" in a PRAM model. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the cases the master theorem solves and under which assumption the theorem holds. Apply then to the binary search algorithm. Is the master theorem of any help in case we would like to estimate the spatial complexity of an algorithm? (2)

B. Please explain following theorem and why does not hold for the counting sort algorithm: Theorem: "Any decision tree that can sort n elements must have height $W(n \lg n)$" (2)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain which optimizations have been considered to reduce the complexity of the disjoint-set operations. (2) (You can skip this question if you completed the first challenge)

B. Describe the treap data structure and how the split is performed, and which complexity has. (2)

Exercise n. 4

Answer the following questions about the amortized analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Detail the Move to Front algorithm. What approach is followed to estimate the temporal complexity of this class of algorithms? Which class of algorithms does it belongs to? (2)

B. What is meant by "potential method"? At which class of algorithms could be applied? Examples? (2)